

TASKS

Question: 1

Measure your CPU temperature at idle and under load using a tool like Monitor or Core Temp, then record the readings before and after applying new thermal paste and reinstalling your CPU cooler?

Answer: I measured the CPU temperature using Core Temp before and after applying new thermal paste and reinstalling the CPU cooler.

Condition	Before Thermal Paste	After Thermal Paste
Idle	48° C	40° C
Under Load	82° C	70° C

After applying new thermal paste and reinstalling the CPU cooler, the CPU temperature decreased in both idle and load conditions. This shows that proper thermal paste application and cooler installation improve heat transfer and CPU cooling performance.

Question:2

Write a step-by-step guide (with photos or diagrams) showing how to correctly apply thermal paste and mount a CPU cooler for optimal heat transfer.

Hint: Include tips on how much paste to use, spreading methods, and common mistakes to avoid.

- Answer:
1. Turn off the computer and unplug it.
 2. Remove the CPU cooler and clean the old thermal paste.
 3. Apply a small pea-sized amount of thermal paste in the center of the CPU.
 4. Place the CPU cooler properly over the CPU.
 5. Tighten the cooler screws evenly in a cross pattern.
 6. Connect the cooler fan to the CPU_FAN header.
 7. Turn on the computer and check the CPU temperature.

Do not use too much paste, leave the surface dirty, or forget to connect to the CPU fan.

Question: 3

Calculate the recommended PSU wattage for three different PC builds: a basic office PC, a gaming PC, and a content creator PC (with GPU and multiple drives). List all major components and show your calculations for each build?

Answer: Basic Office PC

Component	Power
CPU	65W
Motherboard	50W
RAM	10W
SSD	5W
Fans/DSB	30W
Total	160W

Recommended PSU: $160 \times 1.3 = 208W \rightarrow 250W$ PSU

2. Gaming PC

Component	Power
CPU	120W
GPU	250W
Motherboard	60W
RAM	20W
SSD/HDD	15W
Fans/USB	35W
Total	500W

Recommended PSU: $500 \times 1.3 = 650W$ PSU

3. Content Creator PC

Component	Power
CPU	170W
GPU	300W

Motherboard	70W
RAM	40W
2 SSDs + 2 HDDs	40W
Fans	40W
Total	660W

Recommended PSU: $660 \times 1.3 = 858W \rightarrow 900W$ PSU

Question: 4

Compare the total 5-year electricity cost for two PSUs: one rated 80 Plus Bronze (85% efficiency) and one rated 80 Plus Gold (90% efficiency), both running a 400W average load for 8 hours daily at ₹8 per kWh. Show your math and final cost difference. **Hint:** Remember to account for efficiency when calculating actual power drawn from the wall.

Answer: 1. 80 Plus Bronze (85% efficiency):

Power from wall = $400 \div 0.85 = 470.59 \text{ W} = 0.47059 \text{ kW}$

Daily energy = $0.47059 \times 8 = 3.7647 \text{ kWh}$

5-year energy = $3.7647 \times 365 \times 5 = 6,877.46 \text{ kWh}$

Total cost = $6,877.46 \times ₹8 = ₹55,020$

2. 80 Plus Gold (90% efficiency):

Power from wall = $400 \div 0.90 = 444.44 \text{ W} = 0.44444 \text{ kW}$

Daily energy = $0.44444 \times 8 = 3.5556 \text{ kWh}$

5-year energy = $3.5556 \times 365 \times 5 = 6,488.89 \text{ kWh}$

Total cost = $6,488.89 \times ₹8 = ₹51,911$

Question: 5

Create a table comparing air cooling vs liquid cooling for desktop PCs, listing at least 3 pros and 3 cons for each based on performance, noise, maintenance, and cost?

Answer: Feature

Air Cooling

Liquid Cooling

Pros	Low cost; easy Installation; low Maintenance	better cooling performance; quieter under load; good for High - end CPUs
Cons	Can be noisier; less Effective for very Heat; large coolers take Space.	More expensive; difficult installation; pump/leak risk and maintenance